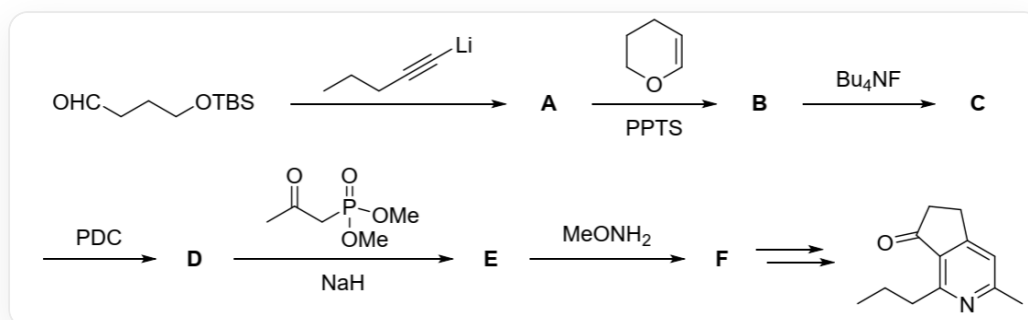


Question

A certain red protoleaf alkaloid with a pentacyclic structure was synthesized, and the construction of two of its rings can be achieved via the following synthetic route:



This is a flowchart-type image containing multiple organic chemical structures and reaction steps, illustrating a reaction pathway proceeding from left to right and top to bottom. The starting molecule in the top-left corner has the SMILES formula C[Si](OCCCC=O)(C)C(C)(C)C. This molecule reacts under arrow-indicated conditions with a substance (SMILES: CCCC#C[Li]), yielding product A. A then proceeds to the right under the reaction conditions "PPTS" and a six-membered oxygen-containing ring (SMILES: C1COC=CC1), generating intermediate B. B undergoes another arrow-directed reaction with the condition "Bu4NF" to form C. From C, the flowchart continues from the lower starting point, where C reacts leftward under the condition "PDC," producing D. D further reacts leftward with a new reactant (SMILES: "CC(CP(OC)(OC)=O)=O") under the condition "NaH," yielding E. E then reacts with "MeONH2" to form F, which proceeds via a pair of bidirectional arrows to the final product on the right, with the SMILES structure CC1=NC(CC)=C2C(CCC2=O)=C1. The image contains no title, axes, legend, units, or numerical labels.

According to the above description, select the option that best fits the question.

- A. Due to the affinity between silicon and oxygen, the TBS group can be efficiently removed in the second step even in the absence of PPTS.
- B. Although silicon and oxygen have good affinity, the second step reaction must be carried out in the presence of PPTS to efficiently remove the TBS group.

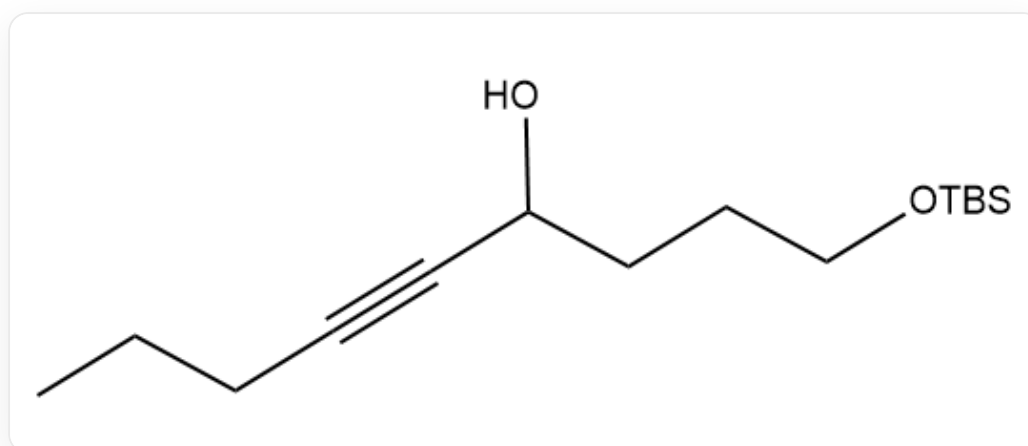
- C.** In the first step of the reaction, the organometallic reagent abstracts the alpha hydrogen of the aldehyde group, efficiently promoting a bimolecular condensation, making the construction of macrocyclic products highly facile.
- D.** In the reaction from **C** to **D**, we efficiently oxidize a secondary hydroxyl group to a ketone.
- E.** In the reaction from **C** to **D**, we efficiently reduce a ketone to a secondary hydroxyl group.
- F.** In the reaction from **E** to **F**, which is a complex one-step process, the synthesis innovatively employs a one-pot method to obtain the product in a single step.
- G.** The purpose of using MeONH_2 is to facilitate potential future Beckmann rearrangement.
- H.** The purpose of using MeONH_2 is to mediate the condensation reaction with carbonyl groups in the future to perform cyclization operations.
- I.** None of the above options is correct.
- J.** Exactly three of the above options are correct.
- K.** Among the above options, exactly two are correct.

Answer

Correct Answer: **I**

Detailed Explanation

Starting material → **A**: The starting material is 4-(tert-butyldimethylsilyloxy)butanal. The reactant is hex-1-ynyllithium. This is a typical nucleophilic addition reaction, where the ynyl anion acts as a nucleophile attacking the carbonyl carbon of the aldehyde group. Product A is a secondary alcohol, and its structure is shown below:



SMILES: CCCC#CC(CCCO[Si](C(C)C)(C(C)C)C(C)C)O

CHECKPOINT

0.5 PTS

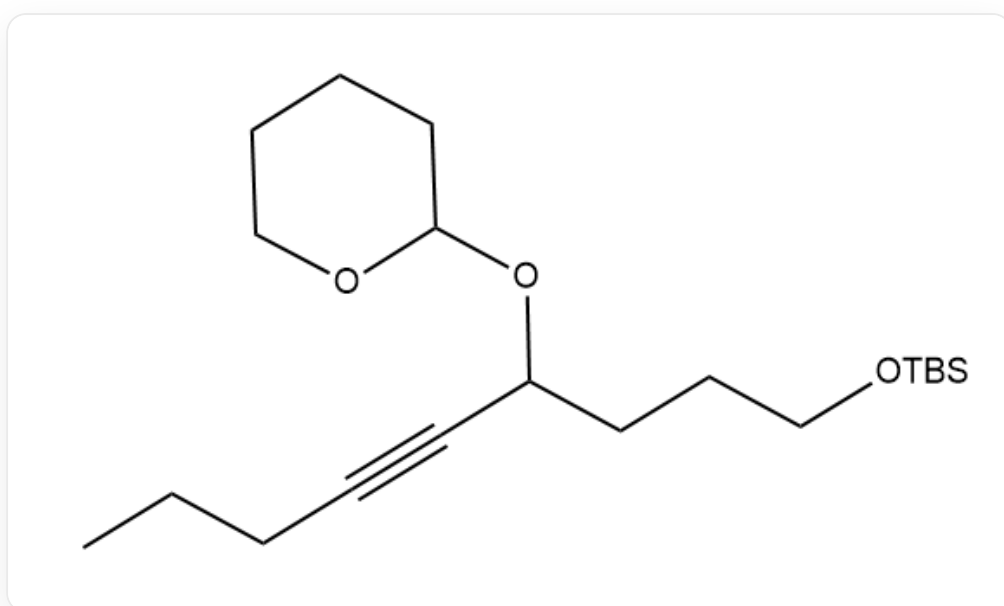
The SMILES structure of **A** is CCCC#CC(CCCO[Si](C(C)C)(C(C)C)C(C)C)O

CHECKPOINT

0.5 PTS

The first step is nucleophilic addition by the ynyl lithium

A → **B**: The reactants are **A**, dihydropyran, and the catalyst PPTS (pyridinium p-toluenesulfonate). PPTS is a mild acid catalyst suitable for THP protection in the presence of the acid-sensitive TBS group. Product **B** is a molecule with both hydroxyl groups protected, as shown below:



SMILES: CCCC#CC(CCCO[Si](C)(C)C(C)(C)C)OC1CCCCO1

CHECKPOINT

0.5 PTS

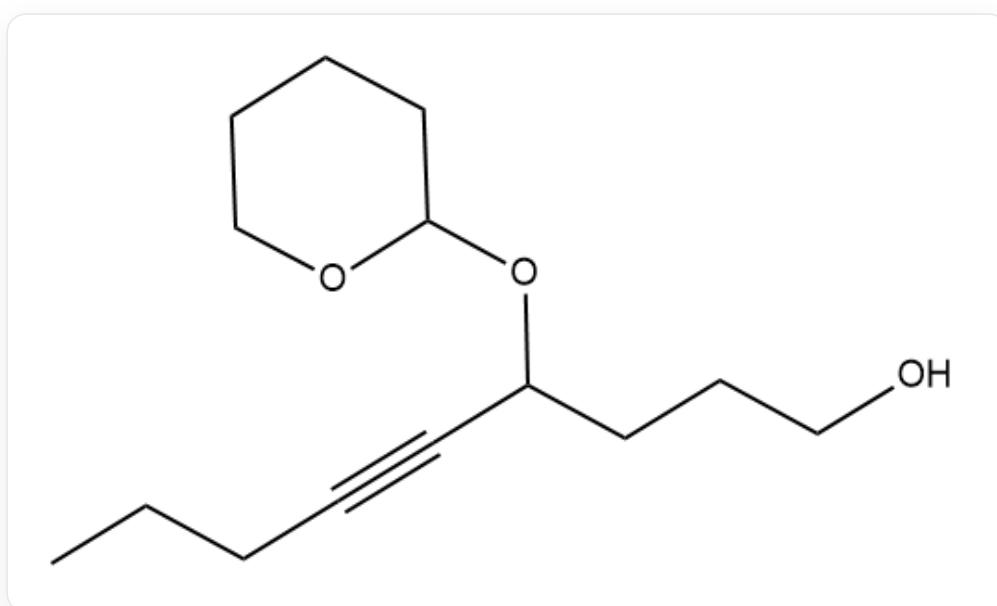
The SMILES structure of **B** is CCCC#CC(CCCO[R5])OC1CCCCO1

CHECKPOINT

0.5 PTS

The second step involves protection using dihydropyran

B → **C**: The reactants are **B** and Bu₄NF (tetrabutylammonium fluoride, TBAF). Due to the strong affinity of fluoride ions for silicon atoms, this step selectively removes the TBS protecting group from the primary alcohol, yielding product **C** as shown below:



SMILES: CCCC#CC(CCCO)OC1CCCCO1

CHECKPOINT

0.5 PTS

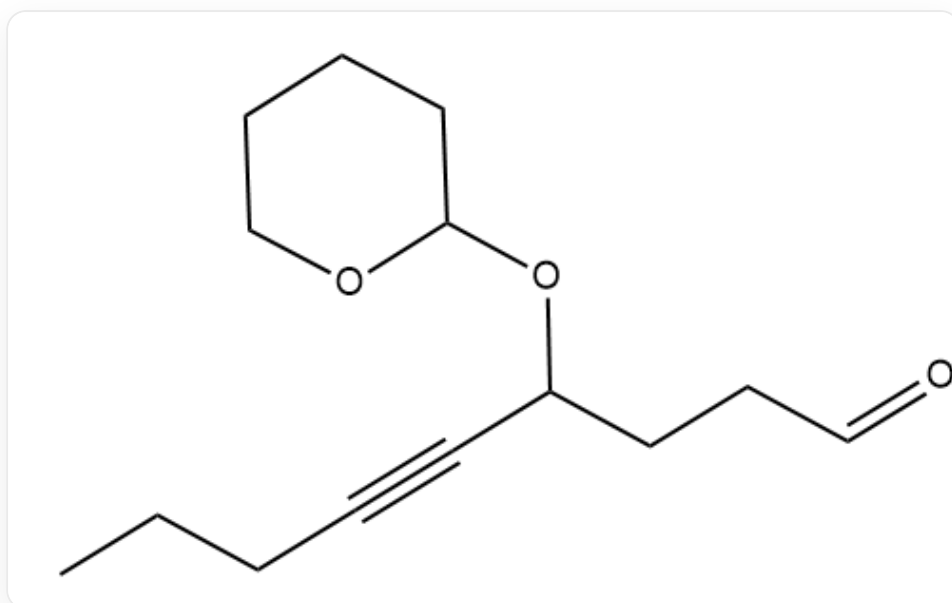
The SMILES structure of **C** is CCCC#CC(CCCO)OC1CCCCO1

CHECKPOINT

0.5 PTS

The third step is TBS deprotection

C → **D**: The reactants are **C** and PDC (pyridinium dichromate). PDC is a moderately strong oxidant that converts primary alcohols to aldehydes without further oxidation to carboxylic acids (in aprotic solvents like CH₂Cl₂). Thus, product **D** is obtained:



CCCC#CC(OC1CCCCO1)CCC=O

CHECKPOINT

0.5 PTS

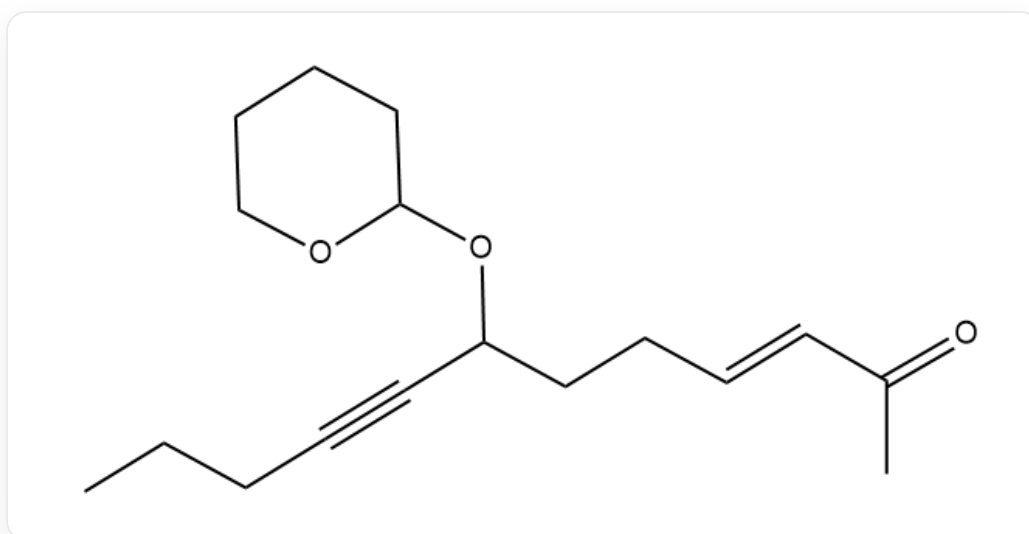
The SMILES structure of **D** is: CCCC#CC(OC1CCCCO1)CCC=O

CHECKPOINT

0.5 PTS

The fourth step is oxidation of the primary alcohol to an aldehyde

D → **E**: The reactants are **D**, a phosphonate (Horner-Wadsworth-Emmons reagent, specifically dimethyl (1-acetyethyl)phosphonate), and the strong base NaH (sodium hydride). This is an HWE olefination reaction. NaH deprotonates the α -position of the phosphonate to form a carbanion, which attacks the aldehyde group in **D**, followed by elimination to yield an α,β -unsaturated ketone. The structure of product **E** is as follows:


CCCC#CC(OC1CCCCO1)CC/C=C/C(C)=O

CHECKPOINT

0.5 PTS

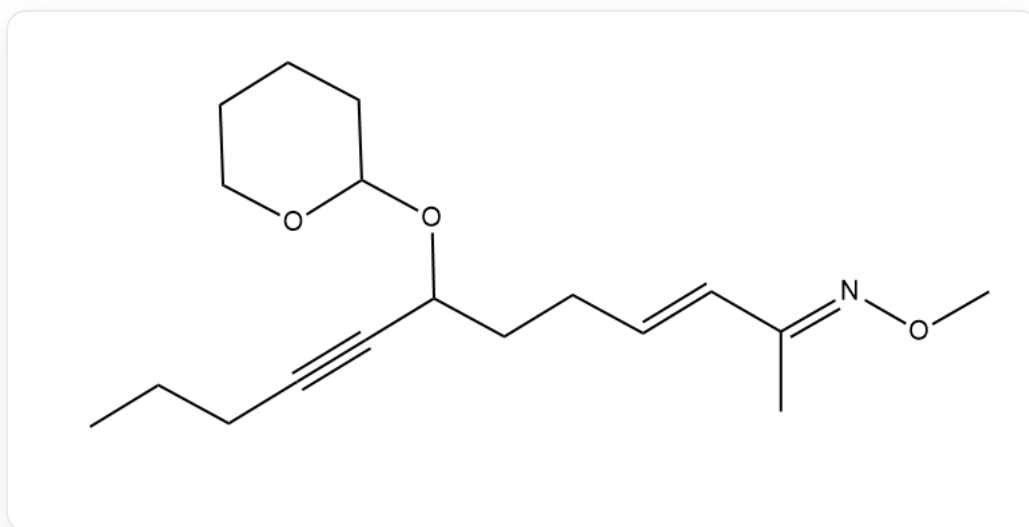
The SMILES structure of **E** is: CCCC#CC(OC1CCCCO1)CC/C=C/C(C)=O

CHECKPOINT

0.5 PTS

The fifth step is the Horner-Wadsworth-Emmons condensation reaction

E → **F**: The reactants are **E** and methoxyamine (MeONH₂). This is a condensation reaction between the aldehyde and methoxyamine, forming an O-methyl oxime. The structure of product **F** is as follows:


CCCC#CC(OC1CCCCO1)CC/C=C/C(C)=N/OC

CHECKPOINT

0.5 PTS

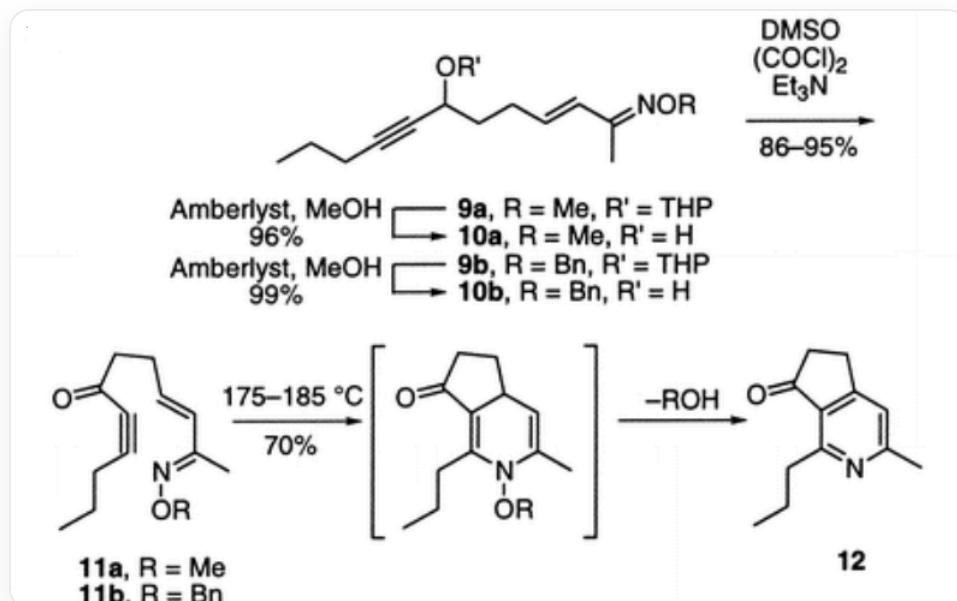
The SMILES structure of **F** is: CCCC#CC(OC1CCCCO1)CC/C=C/C(C)=N/OC

CHECKPOINT

0.5 PTS

The sixth step is the condensation reaction between the aldehyde and methoxyamine

F → Final product: After deprotection, the intermediate strong group is oxidized to a ketone, followed by a DA reaction + elimination to construct the bicyclic structure in one step.



F is first deprotected under Amberlyst, MeOH conditions to give the intermediate secondary hydroxyl group (SMILES: CCCC#CC(O)CC/C=C/C(C)=N/OC), then oxidized to a ketone using DMSO, oxalyl chloride, and triethylamine (SMILES: CCCC#CC(CC/C=C/C(C)=N/OC)=O), and finally reacted at 170–200 °C (triisopropylbenzene solvent, boiling point 233–236 °C). The cycloaddition product intermediate eliminates methanol and is oxidized to the corresponding pyridine, yielding a 70% final product.

CHECKPOINT

1 PTS

The purpose of introducing methoxyamine MeONH2 is to facilitate methanol elimination after the DA reaction